



Nonlinear dynamics and control of space vehicles with multiple fuel slosh modes

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ABSTRACT

This paper studies the modeling and control problem for planar maneuvering of space vehicles with fuel slosh dynamics. A multi-mass–spring model is considered for the characterization of the most prominent sloshing modes. The control inputs are defined by the gimbal deflection angle of a non-throttled thrust engine and a pitching moment about the center of mass of the spacecraft. The control objective, as is typical in a thrust vector control design for a liquid upper stage spacecraft during orbital maneuvers, is to control the translational velocity vector and the attitude of the spacecraft, while attenuating the sloshing modes characterizing the internal dynamics. Subsequently, a nonlinear mathematical model that reflects these specifications is derived. Finally, Lyapunov-based nonlinear feedback control laws are designed to achieve the control objective. A simulation example is included to illustrate the effectiveness of the control laws.

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1. Introduction

Propellant slosh has been a problem studied in spacecraft design since the early days of large, liquid–fuel rockets. In launch vehicles or spacecraft, sloshing can be induced by propellant tank motions resulting from guidance and control system commands or from changes in vehicle acceleration. When the fuel tanks are only partially filled, large quantities of fuel move inside the tanks under translational and rotational accelerations and generate the slosh dynamics. The slosh dynamics interacts with the rigid body dynamics of the spacecraft.

The traditional treatment of liquid slosh control began with the inclusion of physical barriers, such as baffles and complete compartmentalization, meant to limit the movement of liquid fuel to small amplitudes of high, negligible frequencies. Later, bladders were added to the list of ways to limit these motions. These techniques, although helpful in some cases, do not completely succeed in cancelling the sloshing effects. Moreover, these suppression methods involve adding to the spacecraft structural mass, thereby increasing mission cost. Hence the control system must both assure stability during the thrusting phase and achieve good attitude control while suppressing the slosh dynamics.

The effects of baffle positions (and quantities) on sloshing frequency have been studied in the literature (Biswal, Bhattacharyya, & Sinha, 2003). The mathematical techniques used

in these studies are based on the velocity potential function solved using finite-element analysis. Results show that baffles are more effective when near the free-surface of the fluid. In Venugopal and Bernstein (1996), surface pressure control and surface flap actuators have been proposed for controlling slosh in rectangular tanks. The feedback controllers are designed using a Linear-Quadratic-Gaussian (LQG) synthesis. Fluid is assumed to be incompressible, inviscid, and irrotational. Results show a steady-state slosh amplitude lower than the no-actuator case.

The effect of liquid fuel slosh on spinning spacecraft has also been explored in the literature (Hubert, 2003, 2004). Different slosh motion types – surface waves, bulk fluid motion, and vortices – as well as fluid configurations during spinning are defined (Hubert, 2003). The design of control strategies for a launch vehicle with propellant sloshing has also been studied in several works (Blackburn & Vaughan, 1971; Freudenberg & Morton, 1992; Hubert, 2004). In Blackburn and Vaughan (1971), an advanced linear model of the Saturn V launch vehicle is developed and a linear optimal control law is proposed to control the vehicle. The work in Freudenberg and Morton (1992) studies the problem of robust control of a launch vehicle subject to aerodynamic, flexible, and slosh mode instabilities.

It has been demonstrated that pendulum and mass–spring models can approximate complicated fluid and structural dynamics; such models have formed the basis for many studies on dynamics and control of space vehicles with fuel slosh (Bandyopadhyay, Gandhi, & Kurode, 2009a, 2009b; Peterson, Crawley, & Hansman, 1989; Shekhawat, Nickkawde, & Ananthkrishnan, 2006). There is an extensive body of literature on the interaction of vehicle dynamics and slosh dynamics and

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their control, but this literature treats only the case of small perturbations to the vehicle dynamics. The control approaches developed for accelerating space vehicles have commonly employed methods of linear control design (Sidi, 1997; Wie, 1998) and adaptive control (Adler, Lee, & Saugen, 1991). A number of related papers following a similar approach are motivated by robotic systems moving liquid filled containers (Feddema et al., 1997; Grundelius, 2000; Grundelius & Bernhardsson, 1999; Terashima & Schmidt, 1994); (Yano, Higashikawa, & Terashima, 2001); (Yano, Toda, & Terashima, 2001); (Yano & Terashima, 2001). The linear control laws for the suppression of the slosh dynamics inevitably lead to excitation of the transverse vehicle motion through coupling effects. The complete nonlinear dynamics formulation in this paper allows simultaneous control of the transverse, pitch, and slosh dynamics.

The previous work in Cho, McClamroch, and Reyhanoglu (2000b) and Reyhanoglu (2003) considered a spacecraft with a partially filled spherical fuel tank and included only the lowest frequency slosh mode in the dynamic model using pendulum and mass-spring analogies. In this paper, the previous results are extended by using a multi-mass-spring model for the characterization of the most prominent sloshing modes. The control inputs are defined by the gimbal deflection angle of a non-throttleable thrust engine and a pitching moment about the center of mass of the spacecraft. The control objective is to control the translational velocity vector and the attitude of the spacecraft, while attenuating the sloshing modes characterizing the internal dynamics. The results are applied to the AVUM upper stage—the fourth stage of the European launcher Vega (Perez, 2006). The main contributions in this paper are (i) the development of a full nonlinear mathematical model for planar maneuvering of the spacecraft and (ii) the design of Lyapunov-based nonlinear feedback control laws. A simulation example is included to illustrate the effectiveness of the controllers. A preliminary version of this paper was presented as a conference paper (Reyhanoglu & Hervas, 2011).

2. Model formulation

This section formulates the dynamics of a spacecraft with a single propellant tank including the prominent fuel slosh modes. The spacecraft is represented as a rigid body (base body) and the sloshing fuel masses as internal bodies. The main ideas in Cho, McClamroch, and Reyhanoglu (2000a) are employed to express the equations of motion in terms of the spacecraft translational velocity vector, the angular velocity, and the internal (shape) coordinates representing the slosh modes.

To summarize the formulation in Cho et al. (2000a), let $\mathbf{v} \in \mathbb{R}^3$, $\boldsymbol{\omega} \in \mathbb{R}^3$, and $\boldsymbol{\eta} \in \mathbb{R}^N$ denote the base body translational velocity vector, the base body angular velocity vector, and the vector of internal coordinates, respectively. In these coordinates, the Lagrangian has the form $L = L(\mathbf{v}, \boldsymbol{\omega}, \boldsymbol{\eta}, \dot{\boldsymbol{\eta}})$, which is $SE(3)$ -invariant in the sense that it does not depend on the base body position and attitude. The generalized forces and moments on the spacecraft are assumed to consist of control inputs which can be partitioned into two parts: $\boldsymbol{\tau}_t \in \mathbb{R}^3$ (typically from thrusters) is the vector of generalized control forces that act on the base body and $\boldsymbol{\tau}_r \in \mathbb{R}^3$ (typically from symmetric rotors, reaction wheels, and thrusters) is the vector of generalized control torques that act on the base body. It is also assumed that the internal dissipative forces are derivable from a Rayleigh dissipation function R . Then, the equations of motion of the spacecraft with internal dynamics are shown to be given by (Cho et al., 2000a):

$$\frac{d}{dt} \frac{\partial L}{\partial \mathbf{v}} + \hat{\boldsymbol{\omega}} \frac{\partial L}{\partial \mathbf{v}} = \boldsymbol{\tau}_t, \quad (1)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \boldsymbol{\omega}} + \hat{\boldsymbol{\omega}} \frac{\partial L}{\partial \boldsymbol{\omega}} + \hat{\mathbf{v}} \frac{\partial L}{\partial \mathbf{v}} = \boldsymbol{\tau}_r, \quad (2)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\boldsymbol{\eta}}} - \frac{\partial L}{\partial \boldsymbol{\eta}} + \frac{\partial R}{\partial \dot{\boldsymbol{\eta}}} = 0, \quad (3)$$

where $\hat{\mathbf{a}}$ denotes a 3×3 skew-symmetric matrix formed from $\mathbf{a} = [a_1, a_2, a_3]^T \in \mathbb{R}^3$:

$$\hat{\mathbf{a}} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}.$$

It must be pointed out that in the above formulation it is assumed that no control forces or torques exist that directly control the internal dynamics. The objective is to simultaneously control the rigid body dynamics and the internal dynamics using only control effectors that act on the rigid body; the control of internal dynamics must be achieved through the system coupling. In this regard, Eqs. (1)–(3) model interesting examples of underactuated mechanical systems. The published literature on the dynamics and control of such systems includes the development of theoretical controllability and stabilizability results for a large class of systems using tools from nonlinear control theory (Reyhanoglu, van der Schaft, McClamroch, & Kolmanovsky, 1996, 1999) and the development of effective nonlinear control design methodologies (Reyhanoglu, Cho, & McClamroch, 2000) that are applied to several practical examples, including underactuated space vehicles (Cho et al., 2000b; Reyhanoglu, 2003).

In the subsequent section, a mechanical-analogy model is developed to characterize the propellant sloshing during a typical thrust vector control maneuver. The spacecraft acceleration due to the main engine thrust is assumed to be large enough so that surface tension forces do not significantly affect the propellant motion during main engine burns. This situation corresponds to a “high-acceleration” regime that can be characterized by using the Bond number Bo —the ratio of acceleration related forces to the liquid propellant’s surface tension forces, which is given by

$$Bo = \frac{\rho a \mathcal{R}^2}{\sigma},$$

where ρ and σ denote the liquid propellant’s density and surface tension, respectively; a is the spacecraft acceleration, and $\mathcal{R}$ is a characteristic dimension (e.g. propellant tank radius). During the steady-state high-acceleration situation the propellant settles at the “bottom” of the tank with a flat free surface. When the main engine operation for thrust vector control introduces lateral accelerations, the propellant begins sloshing. As discussed in Enright and Wong (1994), Bond numbers as low as 100 would indicate that low-gravity (i.e. low-acceleration) effects may be of some significance. A detailed discussion of low-gravity fluid mechanics is given in Dodge (2000).

2.1. Multi-mass-spring analogy

This section derives a multi-mass-spring model of the sloshing fuel where the oscillation frequencies of the mass-spring elements represent the prominent sloshing modes (Sidi, 1997).

Consider a rigid spacecraft moving on a plane as indicated in Fig. 1, where v_x, v_z are the axial and transverse components, respectively, of the velocity of the center of the fuel tank, and θ denotes the attitude angle of the spacecraft with respect to a fixed reference. The fluid is modeled by moment of inertia I_0 assigned to a rigidly attached mass m_0 and point masses m_i , $i = 1, \dots, N$, whose relative positions along the spacecraft fixed z -axis are denoted by s_i . Moments of inertia I_i of these masses are usually negligible. The locations h_0 and h_i are referenced to the center of

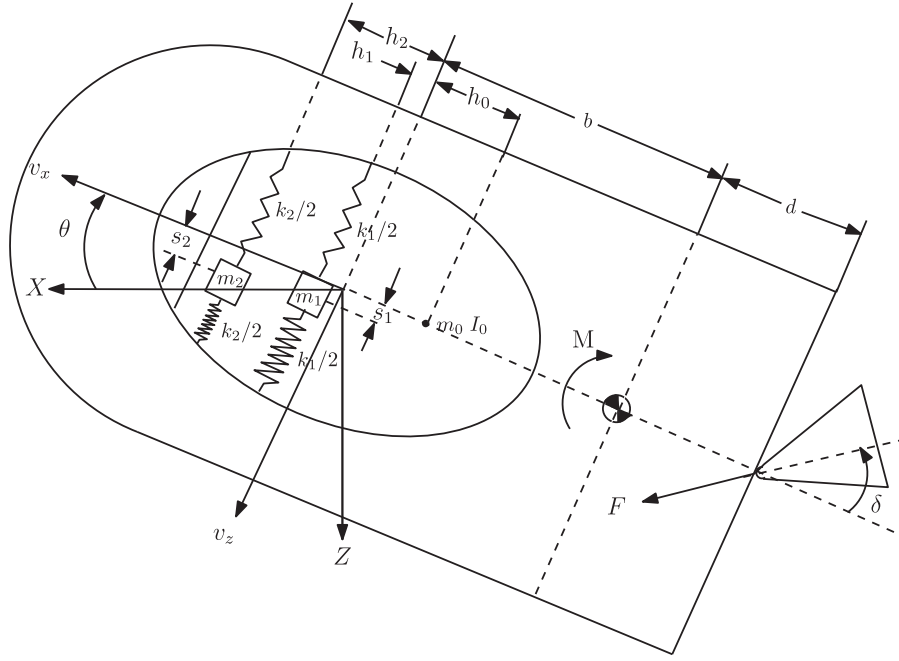


Fig. 1. A multiple slosh mass-spring model for a spacecraft.

the tank. A restoring force $-k_i s_i$ acts on the mass m_i whenever the mass is displaced from its neutral position $s_i=0$. A thrust F is produced by a gimbaled thrust engine as shown in Fig. 1, where δ denotes the gimbal deflection angle, which is considered as one of the control inputs. A pitching moment M is also available for control purposes. The constants in the problem are the spacecraft mass m and moment of inertia I ; the fuel masses m_0, m_i ; the distance b between the body z -axis and the spacecraft center of mass location along the longitudinal axis, and the distance d from the gimbal pivot to the spacecraft center of mass. If the tank center is in front of the spacecraft center of mass then $b > 0$. The parameters m_i, h_i, k_i and b depend on the shape of the fuel tank, the characteristics of the fuel and the fill ratio of the fuel tank. To preserve the static properties of the liquid, the sum of all the masses must be the same as the fuel mass m_f , and the center of mass of the model must be at the same elevation as that of the fuel, i.e.

$$m_0 + \sum_{i=1}^N m_i = m_f, \quad (4)$$

$$m_0 h_0 + \sum_{i=1}^N m_i h_i = 0. \quad (5)$$

Let $\hat{i}$ and $\hat{k}$ be the unit vectors along the spacecraft-fixed longitudinal and transverse axes, respectively, and denote by (x, z) the inertial position of the center of the fuel tank. The position vector of the center of mass of the vehicle can then be expressed in the spacecraft-fixed coordinate frame as

$$\vec{r} = (x-b)\hat{i} + z\hat{k}.$$

Clearly, the inertial velocity of the vehicle can be computed as

$$\dot{\vec{r}} = v_x \hat{i} + (v_z + b\dot{\theta}) \hat{k},$$

where $v_x = \dot{x} + z\dot{\theta}$ and $v_z = \dot{z} - x\dot{\theta}$.

Similarly, the position vectors of the fuel masses $m_0, m_i, \forall i$, in the spacecraft-fixed coordinate frame are given, respectively, by

$$\vec{r}_0 = (x+h_0)\hat{i} + z\hat{k},$$

$$\vec{r}_i = (x+h_i)\hat{i} + (z+s_i)\hat{k} \quad \forall i.$$

Assuming h_i are constants, the inertial velocities can be computed as

$$\dot{\vec{r}}_0 = v_x \hat{i} + (v_z - h_0 \dot{\theta}) \hat{k},$$

$$\dot{\vec{r}}_i = (v_x + s_i \dot{\theta}) \hat{i} + (v_z - h_i \dot{\theta} + \dot{s}_i) \hat{k} \quad \forall i.$$

The total kinetic energy can now be expressed as

$$T = \frac{1}{2} m \dot{\vec{r}}^2 + \frac{1}{2} m \dot{\vec{r}}_0^2 + \frac{1}{2} \sum_{i=1}^N (m_i \dot{\vec{r}}_i^2 + I_i \dot{\theta}^2) + \frac{1}{2} (I + I_0) \dot{\theta}^2.$$

Since gravitational effects are ignored, there is no gravitational potential energy. The acceleration of the rocket gives rise to the elastic potential energy given by

$$U = \frac{1}{2} \sum_{i=1}^N k_i s_i^2.$$

Typically, k_i would be a function of the axial acceleration. However, in this paper it is assumed that the axial acceleration is not significantly affected by small gimbal deflections, pitch changes and fuel motion so that k_i remain constant.

Thus, under the indicated assumptions, the Lagrangian ($L = T - U$) can be computed as

$$L = \frac{1}{2} m [v_x^2 + (v_z + b\dot{\theta})^2] + \frac{1}{2} m_0 [v_x^2 + (v_z - h_0 \dot{\theta})^2] + \frac{1}{2} (I + I_0) \dot{\theta}^2 + \frac{1}{2} \sum_{i=1}^N m_i [(v_x + s_i \dot{\theta})^2 + (v_z - h_i \dot{\theta} + \dot{s}_i)^2] + \frac{1}{2} \sum_{i=1}^N (I_i \dot{\theta}^2 - k_i s_i^2).$$

Dissipative effects due to fuel slosh are included via damping constants c_i . The damping coefficients for the sloshing masses are usually determined by experimental measurements with partially filled tanks (Dodge, 2000). A fraction of kinetic energy of sloshing fuel is dissipated during each cycle of the motion. When the damping is small, it can be represented accurately by equivalent linear viscous damping. Even with baffles, the damping ratio is seldom greater than about 0.05. It is customary to include the

damping via a Rayleigh dissipation function R given by

$$R = \frac{1}{2} \sum_{i=1}^N c_i \dot{s}_i^2.$$

Applying Eqs. (1)–(3) with

$$\boldsymbol{\eta} = \begin{bmatrix} s_1 \\ \vdots \\ s_N \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_x \\ 0 \\ v_z \end{bmatrix}, \quad \boldsymbol{\omega} = \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix},$$

$$\boldsymbol{\tau}_t = \begin{bmatrix} F \cos \delta \\ 0 \\ F \sin \delta \end{bmatrix}, \quad \boldsymbol{\tau}_r = \begin{bmatrix} 0 \\ M + Fp \sin \delta \\ 0 \end{bmatrix},$$

where $p = b + d$, the equations of motion can be obtained as

$$(m + m_f)a_x + mb\dot{\theta}^2 + \sum_{i=1}^N m_i(s_i\ddot{\theta} + 2\dot{s}_i\dot{\theta}) = F \cos \delta, \quad (6)$$

$$(m + m_f)a_z + mb\ddot{\theta} + \sum_{i=1}^N m_i(\ddot{s}_i - s_i\dot{\theta}^2) = F \sin \delta, \quad (7)$$

$$\hat{I}\ddot{\theta} + \sum_{i=1}^N m_i(s_i a_x - h_i \ddot{s}_i + 2s_i \dot{s}_i \dot{\theta}) + mba_z = M + Fp \sin \delta, \quad (8)$$

$$m_i(\ddot{s}_i + a_z - h_i \ddot{\theta} - s_i \dot{\theta}^2) + k_i s_i + c_i \dot{s}_i = 0 \quad \forall i, \quad (9)$$

where $(a_x, a_z) = (\dot{v}_x + \dot{\theta}v_z, \dot{v}_z - \dot{\theta}v_x)$ are the axial and transverse components of the acceleration of the center of tank, and

$$\hat{I} = I + I_0 + mb^2 + m_0 h_0^2 + \sum_{i=1}^N [m_i(h_i^2 + s_i^2) + I_i].$$

The control objective is to design feedback controllers so that the controlled spacecraft accomplishes a given planar maneuver, that is a change in the translational velocity vector and the attitude of the spacecraft, while suppressing the fuel slosh modes.

3. Nonlinear feedback controller design

This section presents a detailed development of a feedback control law through the model obtained via the multi-mass-spring analogy.

Consider the model of a spacecraft with a gimbaled thrust engine shown in Fig. 1. If the thrust F during the fuel burn is a positive constant, and if the gimbal deflection angle and pitching moment are zero, $\delta = M = 0$, then the spacecraft and fuel slosh dynamics have a relative equilibrium defined by

$$v_z = \bar{v}_z, \quad \theta = \bar{\theta}, \quad \dot{\theta} = 0, \quad s_i = 0, \quad \dot{s}_i = 0 \quad \forall i,$$

where $\bar{v}_z$ and $\bar{\theta}$ are arbitrary constants. Without the loss of generality, the subsequent analysis considers the relative equilibrium at the origin, i.e. $\bar{v}_z = \bar{\theta} = 0$. Note that the relative equilibrium corresponds to the vehicle axial velocity

$$v_x(t) = v_{x_0} + \bar{a}_x t, \quad t \leq t_b,$$

where v_{x_0} is the initial axial velocity of the spacecraft, t_b is the fuel burn time, and

$$\bar{a}_x = \frac{F}{m + m_f}.$$

Now assume the axial acceleration term a_x is not significantly affected by small gimbal deflections, pitch changes and fuel motion (an assumption verified in simulations). Consequently, Eq. (6) becomes:

$$\dot{v}_x + \dot{\theta}v_z = \bar{a}_x. \quad (10)$$

Substituting this approximation leads to the following reduced equations of motion for the transverse, pitch and slosh dynamics:

$$(m + m_f)(\dot{v}_z - \dot{\theta}v_x(t)) + mb\ddot{\theta} + \sum_{i=1}^N m_i(\ddot{s}_i - s_i\dot{\theta}^2) = F \sin \delta, \quad (11)$$

$$\hat{I}\ddot{\theta} + \sum_{i=1}^N m_i(\bar{a}_x s_i - h_i \ddot{s}_i + 2s_i \dot{s}_i \dot{\theta}) + mb(\dot{v}_z - \dot{\theta}v_x(t)) = M + Fp \sin \delta, \quad (12)$$

$$\ddot{s}_i + \dot{v}_z - \dot{\theta}v_x(t) - h_i \ddot{\theta} - s_i \dot{\theta}^2 + \frac{k_i}{m_i} s_i + \frac{c_i}{m_i} \dot{s}_i = 0 \quad \forall i, \quad (13)$$

where $v_x(t)$ is considered as an exogenous input. The subsequent analysis is based on the above equations of motion for the transverse, pitch and slosh dynamics of the vehicle.

Remark 1. If one considers only small vehicle motions about the relative equilibrium at the origin, then the following linearized equations of motion can be obtained:

$$(m + m_f)(\dot{v}_z - \dot{\theta}v_x(t)) + mb\ddot{\theta} + \sum_{i=1}^N m_i \ddot{s}_i = F\delta, \quad (14)$$

$$\hat{I}\ddot{\theta} + \sum_{i=1}^N m_i(\bar{a}_x s_i - h_i \ddot{s}_i) + mb(\dot{v}_z - \dot{\theta}v_x(t)) = M + Fp\delta, \quad (15)$$

$$\ddot{s}_i + \dot{v}_z - \dot{\theta}v_x(t) - h_i \ddot{\theta} + \frac{k_i}{m_i} s_i + \frac{c_i}{m_i} \dot{s}_i = 0 \quad \forall i. \quad (16)$$

The origin of the linearized time varying system (14)–(16) can be made uniformly asymptotically stable by linear state feedback. One of the difficulties in control design is the time variation representing the non-constant axial velocity of the vehicle that appears in the equations. This time variation formally prohibits the use of transfer function concepts, even for the linearized system (14)–(16). In the subsequent development, a Lyapunov-based control design approach is proposed for the nonlinear system (11)–(13) that overcomes this difficulty.

Eliminating $\ddot{s}_i$ in (11) and (12) using (13) yields

$$(m + m_0)(\dot{v}_z - \dot{\theta}v_x(t)) + (mb - m_0 h_0)\ddot{\theta} - \sum_{i=1}^N (k_i s_i + c_i \dot{s}_i) = F \sin \delta, \quad (17)$$

$$(mb - m_0 h_0)(\dot{v}_z - \dot{\theta}v_x(t)) + \left(\hat{I} - \sum_{i=1}^N m_i h_i^2 \right) \ddot{\theta} + N(s_i, \dot{s}_i, \dot{\theta}) = M + Fp \sin \delta, \quad (18)$$

where

$$N(s_i, \dot{s}_i, \dot{\theta}) = \sum_{i=1}^N [(m_i \bar{a}_x + k_i h_i) s_i + h_i c_i \dot{s}_i + 2m_i s_i \dot{s}_i \dot{\theta} - m_i h_i s_i \dot{\theta}^2].$$

Note that the expressions (4) and (5) have been utilized to obtain Eqs. (17) and (18) in the form above.

By defining control transformations from (δ, M) to new control inputs (u_1, u_2) :

$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} m + m_0 & mb - m_0 h_0 \\ mb - m_0 h_0 & \hat{I} - \sum_{i=1}^N m_i h_i^2 \end{bmatrix}^{-1} \begin{bmatrix} F \sin \delta + \sum_{i=1}^N (k_i s_i + c_i \dot{s}_i) \\ M + Fp \sin \delta - N(s_i, \dot{s}_i, \dot{\theta}) \end{bmatrix},$$

the system (11)–(13) can be written as:

$$\dot{v}_z = u_1 + \dot{\theta}v_x(t), \quad (19)$$

$$\ddot{\theta} = u_2, \quad (20)$$

$$\ddot{s}_i = -\omega_i^2 s_i - 2\zeta_i \omega_i \dot{s}_i - u_1 + h_i u_2 + s_i \dot{\theta}^2 \quad \forall i, \quad (21)$$

where

$$\omega_i^2 = \frac{k_i}{m_i}, \quad 2\zeta_i \omega_i = \frac{c_i}{m_i} \quad \forall i.$$

Here ω_i and ζ_i , $\forall i$, denote the undamped natural frequencies and damping ratios, respectively.

The main idea in the subsequent development is to first design feedback control laws for (u_1, u_2) and then use the following equations to obtain the feedback laws for the original controls (δ, M) for $t \leq t_b$:

$$\delta = \sin^{-1} \left(\left[(m + m_0)u_1 + (mb - m_0 h_0)u_2 - \sum_{i=1}^N (k_i s_i + c_i \dot{s}_i) \right] / F \right), \quad (22)$$

$$M = (mb - m_0 h_0)u_1 + \left(\hat{I} - \sum_{i=1}^N m_i h_i^2 \right) u_2 + N(s_i, \dot{s}_i, \dot{\theta}) - Fp \sin \delta. \quad (23)$$

Now, consider the following candidate Lyapunov function for the system (19)–(21):

$$V = \frac{r_1}{2} v_z^2 + \frac{r_2}{2} \theta^2 + \frac{r_3}{2} \dot{\theta}^2 + \frac{r_4}{2} \sum_{i=1}^N (\dot{s}_i^2 + \omega_i^2 s_i^2 - 2h_i \dot{s}_i \dot{\theta}),$$

where r_1, r_2, r_3 , and r_4 are positive constants. Assume that

$$\mu = r_3 - r_4 \sum_{i=1}^N h_i^2 > 0$$

so that the function V is positive definite.

Remark 2. Let $\mathbf{z} = (\dot{\theta}, \dot{s}_1, \dots, \dot{s}_N)^T$ and let $\mathbf{Q} \in \mathbb{R}^{(N+1) \times (N+1)}$ denote the symmetric matrix corresponding to the quadratic form

$$\mathbf{z}^T \mathbf{Q} \mathbf{z} = r_3 \dot{\theta}^2 + r_4 \sum_{i=1}^N (\dot{s}_i^2 - 2h_i \dot{s}_i \dot{\theta}).$$

Clearly, this quadratic form is positive definite if and only if the leading principal minors of the matrix $\mathbf{Q}$ are all positive. It is easy to show that the condition holds if $\mu > 0$.

The time derivative of V along the trajectories of (19)–(21) is

$$\begin{aligned} \dot{V} = & -2r_4 \sum_{i=1}^N \zeta_i \omega_i \dot{s}_i^2 + \left[r_1 v_z - r_4 \sum_{i=1}^N (\dot{s}_i - h_i \dot{\theta}) \right] u_1 + \left[r_1 v_x(t) v_z + r_2 \dot{\theta} + \mu u_2 \right. \\ & \left. + r_4 \sum_{i=1}^N (h_i \omega_i^2 s_i + 2\zeta_i \omega_i h_i \dot{s}_i + s_i \dot{s}_i \dot{\theta} - h_i s_i \dot{\theta}^2) \right] \dot{\theta}. \end{aligned}$$

Clearly, the feedback laws

$$u_1 = -K_1 \left[r_1 v_z - r_4 \sum_{i=1}^N (\dot{s}_i - h_i \dot{\theta}) \right], \quad (24)$$

$$u_2 = -\frac{1}{\mu} \left[r_2 \dot{\theta} + K_2 \dot{\theta} + r_1 v_x(t) v_z + r_4 \sum_{i=1}^N (h_i \omega_i^2 s_i + 2\zeta_i \omega_i h_i \dot{s}_i + s_i \dot{s}_i \dot{\theta} - h_i s_i \dot{\theta}^2) \right], \quad (25)$$

where K_1 and K_2 are positive constants, yield

$$\dot{V} = -K_1 \left[r_1 v_z - r_4 \sum_{i=1}^N (\dot{s}_i - h_i \dot{\theta}) \right]^2 - K_2 \dot{\theta}^2 - 2r_4 \sum_{i=1}^N \zeta_i \omega_i \dot{s}_i^2,$$

which satisfies $\dot{V} \leq 0$.

Note that the closed-loop system becomes time-invariant after the burnout (since the time-varying term $v_x(t)$ becomes constant after the burnout). Using an invariance principle for time-varying systems (Khalil, 2002), it is easy to prove asymptotic stability of

the origin of the closed loop defined by Eqs. (19)–(21) and the feedback control laws (24)–(25). Note also that the positive gains K_1 and K_2 can be chosen arbitrarily to achieve good closed loop responses.

Note also that as mentioned previously the control approaches developed for accelerating space vehicles have commonly employed methods of linear control design. The linear control laws for the suppression of the slosh dynamics inevitably lead to excitation of the transverse vehicle motion through nonlinear coupling effects. In this paper, this issue has been addressed by designing the nonlinear controller given by (22)–(25) based on the complete nonlinear dynamics formulation that allows simultaneous control of the transverse, pitch, and slosh dynamics.

4. Simulation

The feedback control laws developed in the previous section are implemented here for the AVUM upper stage spacecraft (Perez, 2006). The first two slosh modes are included to demonstrate the effectiveness of the control law. The physical parameters used in the simulation are given in Table 1. The fluid parameters are obtained using the formulae in Dodge (2000).

The control objective is the stabilization of the spacecraft in orbital transfer, suppressing the transverse and pitching motion of the spacecraft and sloshing of fuel while the spacecraft is accelerating. In other words, the control objective is to stabilize the relative equilibrium corresponding to a constant axial spacecraft acceleration of 1.77 m/s^2 and $v_z = \theta = \dot{\theta} = 0$, $s_i = \dot{s}_i = 0$, $i = 1, 2$. In the simulation, a fuel burn time of 660 s is assumed.

It must be noted that for the AVUM spacecraft the characteristic length of the propellant tank can be taken as $\approx 0.5 \text{ m}$ and the propellant UMDH (Unsymmetrical Dimethyl Hydrazine) has a σ/ρ ratio of around $0.25 \times 10^{-4} \text{ m}^3/\text{s}^2$. Thus, accelerations that are larger than 0.1 m/s^2 correspond to a Bond number larger than 1000, which is clearly in the high-acceleration regime. The simulation indicates that during the main engine burn the vehicle acceleration exceeds 1 m/s^2 , and thus the mechanical-analogy model is valid.

The effectiveness of the Lyapunov-based control laws is demonstrated by applying the controller defined by (22)–(25) to the complete nonlinear system (6)–(9). Time responses shown in Figs. 2–4 correspond to the initial conditions $v_{x_0} = 3000 \text{ m/s}$, $v_{z_0} = 150 \text{ m/s}$, $\theta_0 = 5^\circ$, $\dot{\theta}_0 = 0$, $s_{1_0} = 0.15 \text{ m}$, $s_{2_0} = -0.15 \text{ m}$, and $\dot{s}_{1_0} = \dot{s}_{2_0} = 0$. As can be seen in the figures, the transverse velocity, attitude angle, and the slosh states converge to the relative equilibrium at zero while the axial velocity v_x increases and $\dot{v}_x$ tends asymptotically to 1.77 m/s^2 . Note that there is a trade-off between good responses for the directly actuated degrees of freedom (the transverse and pitch dynamics) and good responses for the unactuated degree of freedom (the slosh dynamics); the controller given by (22)–(25) with parameters $r_1 = 8 \times 10^{-7}$,

Table 1
Physical parameters.

Parameter	Value	Parameter	Value
m	975 kg	F	2450 N
I	400 kg · m ²	I_0	44.1 kg · m ²
I_1	10 kg m ²	I_2	1 kg · m ²
m_0	205 kg	k_1	1174 N/m
m_1	195 kg	k_2	120 N/m
m_2	6.5 kg	c_1	48 N s/m
h_0	0.135 m	c_2	2.75 N s/m
h_1	−0.145 m	b	−0.6 m
h_2	0.035 m	d	1.2 m

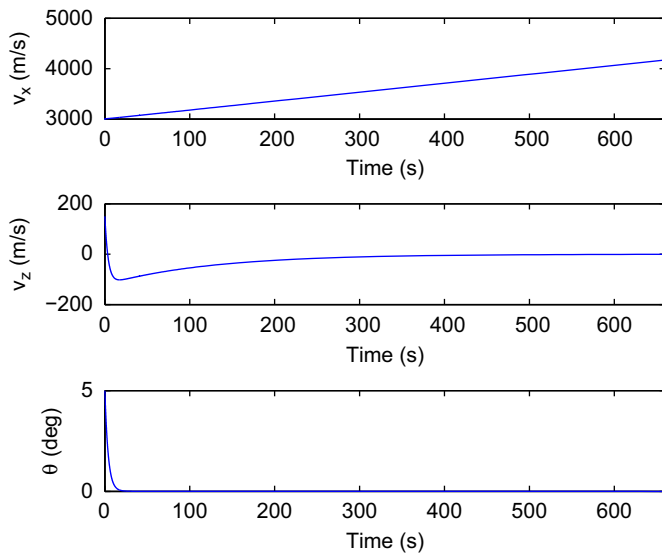


Fig. 2. Time responses of v_x , v_z and θ .

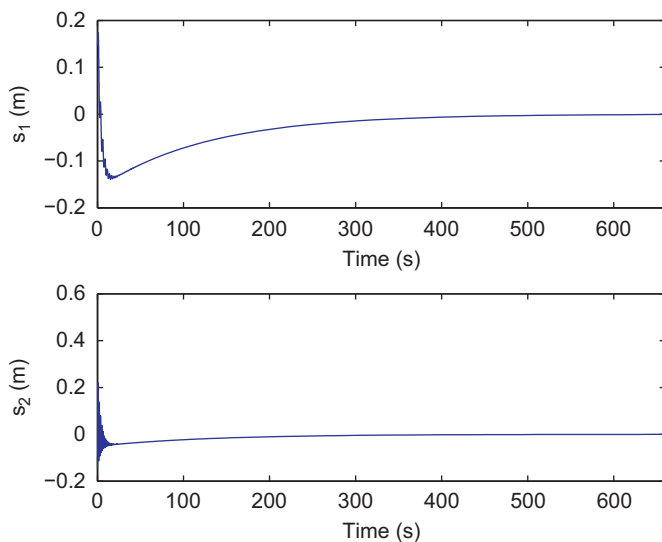


Fig. 3. Time responses of s_1 and s_2 .

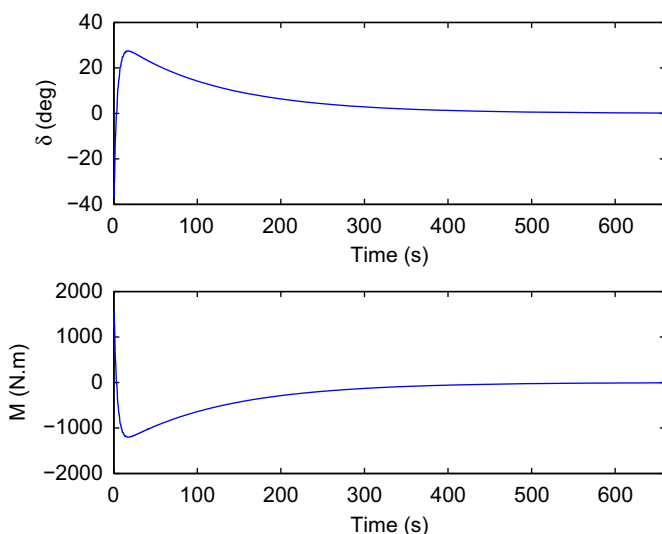


Fig. 4. Gimbal deflection angle δ and pitching moment M .

$r_2 = 2500$, $r_3 = 500$, $r_4 = 1 \times 10^{-5}$, $K_1 = 1 \times 10^4$, $K_2 = 1 \times 10^4$ represents one example of this balance.

5. Conclusions

A complete nonlinear dynamical model has been developed for space vehicles with multiple slosh modes. Lyapunov-based nonlinear feedback control laws have been designed to achieve stabilization of the pitch and transverse dynamics as well as suppression of the slosh modes, while the spacecraft accelerates in the axial direction. The effectiveness of the feedback control laws have been illustrated through a simulation example. The many avenues considered for future research include problems involving multiple propellant tanks and three dimensional maneuvers. Future research also includes designing nonlinear control laws that achieve robustness, insensitivity to system and control parameters, and improved disturbance rejection.

The control system design in this paper is based on the assumption that the axial acceleration term of the spacecraft is not significantly affected by small gimbal deflections, pitch changes and fuel motion so that it remains constant. Moreover, it is assumed that the slosh parameters do not significantly change in time. Relaxing these assumptions presents another promising avenue of research.

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